

Machine Learning

Final Exam

**** WITH ANSWERS ****

23 March 2026

12:15–14:30

The questions start on the next page.

Feel free to write on the exam booklet, and take it with you afterwards. The exam will be made available on Canvas later today, together with the correct answers. **You can copy your final answers onto the booklet, so you can check how you did right away.**

There are 40 questions, worth 1 point each.

To pass the examination, your total points for the exam plus those for the quizzes should be 52 points or more.

Rules:

- You are allowed to use a calculator or graphical calculator.
- You are *not* allowed to use your phone or smartphone.
- The exam is closed-book.
- You are allowed to use the formula sheet provided through Canvas.
- The formula sheet should not have any writing on it, except in the cheat sheet box. Anything inside the box should be written by hand.

Recall questions

1. Which of the following abstract tasks is *supervised*?

- A Clustering
- B Regression ✓
- C Density Estimation
- D Generative Modeling

In regression the target labels are numbers, which are given in the training data. Thus, we supervise the algorithm by showing it what it needs to predict. In clustering, we separate the dataset in clusters without any examples of what kind of clusters we are looking for. In density estimate, we fit a probability distribution to the data. In generative modeling, we also fit a probability distribution, but one that we can only sample from. None of these require examples of the kind of thing we want to assign to an instance in the data.

2. Why do the slides say “sometimes your loss function should not be the same as your evaluation function.”

- A The evaluation function may provide a smooth loss surface.
- B The evaluation function may lead to a loss surface with only flat regions and sharp changes. ✓
- C The evaluation function may not be linear, and a loss function must be linear.
- D The evaluation function may not have a global optimum.

See the picture shown here. The flat regions are the polygons where the loss is constant. The sharp changes are the edges. Gradient descent will not work on such a loss surface. A smooth loss surface is exactly what we want. There is no need for a loss surface to be linear (this would indeed be unusual, since linear functions don't have a minimum (or a minimum at infinity, if you like). The evaluation function could have many or few local optima (this depends more on the model than the loss/evaluation function).

3. In terms of over- and underfitting, what can *feature expansion*—adding features derived from existing features—and *dimensionality reduction* do for us?

- A Both can reduce overfitting.
- B Both can reduce underfitting.
- C Dimensionality reduction can reduce overfitting and feature expansion can reduce underfitting. ✓
- D Dimensionality reduction can reduce underfitting and feature expansion can reduce overfitting.

Feature expansion allows us to use a linear classifier to fit non-linear decision boundaries. That means it makes solves underfitting, but risks overfitting. Dimensionality reduction goes in the opposite direction: it can reduce overfitting, as explained here: <https://mlvu.github.io/preprocessing/#slide-114>

4. We are training a classification model by gradient descent, and we want to figure out which learning rate to use, before comparing the model to other classifiers. We try five learning rate values, resulting in five different models. How do we choose among these five models?

- A We measure the accuracy of each model on the training set.
- B We measure the accuracy of each model on the validation set. ✓
- C We measure the accuracy of each model on the test set.
- D We measure the accuracy of each model on the full dataset.

5. If we have a symbolic sequence, we can encode the elements of the sequence as vectors in two ways: *one-hot* vectors and *embedding* vectors. Which is **true**?

- A If the embedding vectors are multiplied by a weight matrix, the operation is equivalent to using one-hot vectors without the weight matrix.
- B If the embedding vectors are passed through a sigmoid nonlinearity, the operation is equivalent to using one-hot vectors without the nonlinearity.
- C Storing one-hot vectors naively takes more memory than storing embedding vectors. ✓
- D Storing embedding vectors naively takes more memory than storing one-hot vectors.

One-hot vectors and embeddings are explained starting here: <https://mlvu.github.io/sequences/#slide-039>. We can get embedding vectors by multiplying a one-hot vector by a weight matrix (<https://mlvu.github.io/sequences/#slide-044>) but not the other way around. A sigmoid nonlinearity does not turn one into the other.

6. What is **true** about *logistic regression*?

- A Logistic regression is a discriminative classifier with a cross-entropy loss function. ✓
 - B Logistic regression is a generative classifier with a cross-entropy loss function.
 - C Logistic regression is a discriminative classifier with a least-squares loss function.
 - D Logistic regression is a generative classifier with a least-squares loss function.
7. Which is a good strategy to deal with missing labels in a dataset, if they are not missing uniformly?
- A Impute the missing labels in the training and test set.
 - B Train only on labeled data, and impute the missing labels in the test set.
 - C Train only on labeled data, and ignore the missing labels in the test set.
 - D Impute the missing labels in the training set, and report your uncertainty for the test set. ✓
8. In which of the following settings would you use *walk-forward validation*?
- A You want to do classification, but only 10% of your data is labeled.
 - B You observe 10 times as many negatives as positives.
 - C Your instances are ordered in time. ✓
 - D You have a relatively small amount of training data, and you'd like to get good estimates of your validation performance.
- Explained here: <https://mlvu.github.io/evaluation/#slide-024> and here: <https://mlvu.github.io/lecture11/#slide-009>
9. Which of the following is an ensemble method?
- A Q-Learning.
 - B k-Means.
 - C AdaBoost. ✓
 - D All of the above.
- See <https://mlvu.github.io/lecture10/#video-037>.
- Q- Learning is a reinforcement learning method. k-Means is a clustering method.
10. In deep learning, what is one reason for expressing our model computations in tensor operations like matrix multiplication, vector addition and so on?

- A To enable complex data visualization.
- B To speed up computations by parallel processing. ✓
- C To simplify dataset labeling.
- D To reduce model training time through compression.

11. What is a key feature of the Transformer architecture that differentiates it from other sequence-to-sequence models in natural language processing?

- A It uses recurrent layers to process the input sequence.
- B It requires fixed-length input sequences.
- C It relies primarily on attention mechanisms to learn interactions across the time dimension. ✓
- D It uses convolutional layers to capture sequential information.

The Transformer architecture uses self-attention rather than recurrent mechanisms. While it has a maximum context length in practice, it can always process sequences longer or shorter than the context length. In a transformer architecture, using the definition we use in the lecture, the self-attention is the only part that propagates information along the time dimension. See details here: <https://mlvu.github.io/transformers/#slide-023>.

Convolutions are not traditionally a part of the transformer architecture.

12. Which of the following best describes the primary operation of a self-attention layer?

- A It uses recurrent connections to process inputs sequentially.
- B It applies convolution filters to extract local features.
- C It computes each output as a weighted sum of the inputs, where the weights are derived from the inputs themselves. ✓
- D It uses a fixed set of learned weights to combine input elements.

Explained here: <https://mlvu.github.io/transformers/#slide-006>.

13. What do we call the situation where a neural network is initialized so that its sigmoid activations are close to 0 and 1, and the first layers have no gradient to start learning with.

- A Stochastic gradient descent.
- B The cold-start problem.
- C The vanishing gradient problem. ✓
- D High variance.

Described here <https://mlvu.github.io/deeplearning/#slide-038>.

The cold start problem occurs in recommender systems. Network initialization and sequence-to-label learning have nothing to do with it. Answer C is a red herring: it talks about recommender systems, but the language (sampling a rating, latent space) doesn't make sense for recommender systems. Answer D describes the issue in matrix factorization models, which are a subset of recommender systems.

Combination questions

14. Jim and Paula have trained a classifier to detect whether people in photos are male or female. There are twice as many men in the training data as women. Both are worried that the classifier is biased. Jim suggests that high bias can be solved with by using *boosting*. Paula thinks that re-balancing the dataset might help to mitigate this issue. Who is right?
- A Jim
 - B Paula
 - C Both are wrong, because they are using the word bias to mean different things.
 - D Both can be right, but they are using the word bias to mean different things. ✓
15. Why can a machine learning model with high accuracy still have negative social consequences?
- A Because high accuracy implies that the model will overfit to minority groups.
 - B Because high accuracy guarantees that the model will be biased towards the majority class.
 - C Because high accuracy is only possible if the model overfits the training data.
 - D Because high accuracy only measures overall performance, potentially masking poor outcomes for specific groups. ✓
16. A *lazy* algorithm is a machine learning method that simply stores the training data and refers back to it during evaluation, instead of actively training a model which can be stored independently of the data.
- Which of the following methods is a lazy algorithm?

- A A regression tree
- B Adaboost
- C k-Means
- D k-Nearest neighbours ✓

17. We use PCA to decrease the dimensionality of some data. We notice that the first principal component is responsible for more than 99% of the overall variance of the data. What does it tell you about the data?

- A The data lies approximately on a line. ✓
- B The data is likely normally distributed.
- C The data is likely biased.
- D The data likely contains missing values.

18. Sometimes, we want to learn a model that maps some input to some output, but we don't have matched input-output pairs: only sets of examples from the input domain and from the output domain. Which method is designed for this situation?

- A Vanilla GAN
- B Conditional GAN
- C CycleGAN ✓
- D StyleGAN

The vanilla GAN is a pure generator: it just generates things that look like they belong in the training data. The conditional GAN generates with some conditioning information, but still needs input/output pairs. The Cycle GAN employs a trick to make this work even when the inputs and outputs aren't paired. the StyleGAN is a vanilla GAN with a specific generator architecture to help separate style and content.

More details:

19. The standard decision tree algorithm doesn't stop adding nodes until all leaves either contain no data instances, or only instances with the same label (or all features have been used). Why is this a problem, and what is the default solution (mentioned in the slides)?

- A It's a problem because the algorithm may never terminate. To solve it, we can use a validation set to see if removing nodes improves performance.
 - B It's a problem because the algorithm may never terminate. To solve it, we can remove features from the data, so that fewer splits are available.
 - C It's a problem because we may be overfitting. To solve it, we can use a validation set to see if removing nodes improves performance. ✓
 - D It's a problem because we may be overfitting. To solve it, we can add features to the data that are computed from the original features.
20. In a maximum-margin loss, which is an advantage of rewriting the constrained problem into an unconstrained problem?
- A It allows some points to fall inside the margin.
 - B It allows some points to be misclassified.
 - C It allows us to use gradient descent for optimization. ✓
 - D It finds better decision boundaries even for linearly-separable classes.
21. The bias/variance decomposition is a helpful tool. Which is **false**?
- A If we know that we have high bias, we can use ensembling methods to reduce it.
 - B If we know that we have high variance, we can use ensembling methods to reduce it.
 - C We can estimate the bias/variance decomposition of our error if we have multiple datasets, or a way to resample data.
 - D We can compute the bias/variance decomposition of our error for a single model trained on a single dataset. ✓
- High variance can be reduced by bagging. High bias can be reduced by boosting. Bias and variance can only be estimated by repeating the experiment on multiple datasets from the same source. Bootstrapping can be used to simulate this process.
22. There are two pairs of performance metrics that are often used together: precision/recall and true positive rate/false negative rate (ROC space). Which is **true**?
- A These are four different metrics, with different formulas.
 - B The false negative rate is the same as the precision.
 - C The true positive rate is the same as the recall. ✓
 - D You can map one onto the other by a simple rescaling of axes.
23. Sandra has built a linear classifier that detects whether a small snippet of code contains a bug—based on a large set of features she has designed.

When she resamples her data and trains the classifier again, she finds that the performance and the instances the classifier gets wrong, are both always the same. What ensembling method can help Sandra to improve her classifier?

- A L1 regularization
- B Q-learning
- C Bagging
- D Boosting ✓

Only the last two are ensembling methods. Sandra's observation shows that she has high bias and not high variance, so boosting is required.

24. What is the difference between *inductive* and *transductive* learning?
- A In transductive learning the model is allowed to see the labels of the test data during training.
 - B In inductive learning the model is allowed to see the labels of the test data during training.
 - C In transductive learning, the model is allowed to see the features of the test data during training. ✓
 - D In inductive learning, the model is allowed to see the features of the test data during training.

Defined here

25. To train a deep neural network in a reinforcement learning setting, we can use *policy gradients (PG)* or *deep Q-learning (DQN)*. What is one difference between these two methods?
- A In PG, we use the fact that we know the structure of the game tree when learning games like chess and go. In DQN, we don't.
 - B In DQN, we use the fact that we know the structure of the game tree when learning games like chess and go. In PG, we don't.
 - C In PG we train a policy network, but in DQN we train only a value network.
 - D In PG, we predict a probability distribution over the actions given the state. In DQN, we predict a Q-value for every action given the state. ✓

Application questions

We have the following training set:

	x_1	x_2	label
a	4	9	Pos
b	1	8	Pos
c	2	0	Neg
d	3	1	Neg
e	0	2	Pos
f	-1	4	Pos
g	7	3	Neg

For the following questions, it helps to draw the data and the classification boundary in feature space.

We use a linear classifier defined by

$$c(x_1, x_2) = \begin{cases} \text{Pos} & \text{if } 0 \cdot x_2 - x_1 > -2 \\ \text{Neg} & \text{otherwise.} \end{cases}$$

26. If we turn c into a *ranking* classifier, how does it rank the points, from most **Negative** to most **Positive**?

- A a b f g e d c
- B c d e g f b a ✓
- C f e b c d a g
- D g a d c b e f

The simplest way to solve this question is to draw the instances and the decision boundary in feature space. For a shortcut, you can recognize that x_2 is ignored by the classifier, so you only need to draw the one-dimensional space of x_1 . The decision boundary is at $x_1 = 2$. Check carefully on which side the **positives** fall and on which side the **negatives** fall. Then, order the instances along the line from most negative to most positive, as the question asks. If you've drawn the x_1 axis horizontally (increasing from left to right), you can read off the instances from right to left (because the lower x_1 is, the more **positive** the classifier finds it).

27. How many ranking errors does the classifier make?

- A 1
- B 2 ✓
- C 3
- D 4

Note that a *ranking error* is not a single instance for which the classifier makes a mistake: it's a pair of instances, one **negative** and one **positive**, that have been classified **the wrong way around**. That is, c is ranked as more

positive than a even though c is negative and a is positive. The safest way to solve this is to draw the coverage matrix (which you need anyway for the next question). Draw a 4×3 matrix with the positives vertically (abef from top to bottom), the negatives horizontally (cdg from left to right) and the instances ordered with the most positive towards the bottom left corner. Each cell is a pair that could be ranked the wrong way around. Color these red and the rest green. The red cells $((a, c)$ and $(a, d))$ should cluster together at the top left.

28. If we draw a coverage matrix (as done in the slides), what proportion of the cells will be red (that is, marked as wrong)?

A $\frac{1}{12}$ B $\frac{1}{6}$ ✓ C $\frac{1}{4}$ D $\frac{1}{3}$

The size of the coverage matrix is the number of positives times the number of negatives ($4 \times 3 = 12$). The number of red cells is the number of ranking errors (2) so the proportion of red cells is $2/12$.

...

We will use the backpropagation algorithm to find the derivative of the function

$$f(x) = \sin\left(\frac{3}{\cos x}\right) \cos x$$

with respect to x .

First, we break the function up into modules:

$$f = ac$$

$$a = \cos x$$

$$b = 3/a$$

$$c = \sin b$$

29. What should be in the place of the dots for a and c ?

A $f = ac$, $b = 3/\cos a$

B $f = ac$, $b = 3/a$ ✓

C $f = 3a$, $b = 3/\cos a$

D $f = 3a$, $b = 3/a$

The answers above show the correct solution. The simplest way to work this out is just to fill in the given answers and to see if you recover the function given in the question.

30. In terms of the *local* derivatives. Which is the correct expression for the derivative $\frac{\partial f}{\partial x}$?

- A $c \sin x + \frac{3}{a^2} \sin x$
- B $c \sin x - \frac{3}{a^2} \sin x$
- C $-\sin(x) \left(c - \frac{3 \cos b}{a^2} \right)$
- D $-\sin(x) \left(c - \frac{3 \cos b}{a} \right) \checkmark$

First, write down the computation graph. This consists of two paths from $x \rightarrow a \rightarrow b \rightarrow c \rightarrow f$ and $a \rightarrow f$. The multivariate chain rule tells us to take the chain rule over both inputs to f and to sum them (or, equivalently to apply the regular chain rule over both paths, and to sum these). Step by step, we get:

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} + \frac{\partial f}{\partial c} \frac{\partial c}{\partial x} && \text{Apply the multivariate chain rule to } f. \\ &= \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} + \frac{\partial f}{\partial c} \frac{\partial c}{\partial b} \frac{\partial b}{\partial a} \frac{\partial a}{\partial x} && \text{Apply the chain rule to } \partial c / \partial x. \quad (1) \end{aligned}$$

We keep going until we've reduced every factor to the derivative of the output of a module with respect to its inputs.

Next, we work out these *local* derivatives and fill them in. They are:

$$\begin{aligned}\frac{\partial f}{\partial \mathbf{a}} &= \frac{\partial \mathbf{ac}}{\partial \mathbf{a}} = \mathbf{c} \\ \frac{\partial f}{\partial \mathbf{c}} &= \frac{\partial \mathbf{ac}}{\partial \mathbf{c}} = \mathbf{a} \\ \frac{\partial \mathbf{c}}{\partial \mathbf{b}} &= \cos \mathbf{b} \\ \frac{\partial \mathbf{b}}{\partial \mathbf{a}} &= \frac{\partial 3\mathbf{a}^{-1}}{\partial \mathbf{a}} = -3\mathbf{a}^{-2} = -\frac{3}{\mathbf{a}^2} \\ \frac{\partial \mathbf{a}}{\partial \mathbf{x}} &= -\sin \mathbf{x}\end{aligned}$$

Filling these in to the result from line 1 and rewriting, we get

$$\begin{aligned}\frac{\partial f}{\partial \mathbf{x}} &= \frac{\partial f}{\partial \mathbf{a}} \frac{\partial \mathbf{a}}{\partial \mathbf{x}} + \frac{\partial f}{\partial \mathbf{c}} \frac{\partial \mathbf{c}}{\partial \mathbf{b}} \frac{\partial \mathbf{b}}{\partial \mathbf{a}} \frac{\partial \mathbf{a}}{\partial \mathbf{x}} \\ &= \mathbf{c} \times -\sin \mathbf{x} + \mathbf{a} \times \cos \mathbf{b} \times -\frac{3}{\mathbf{a}^2} \times -\sin \mathbf{x} \\ &= -\sin \mathbf{x} \left(\mathbf{c} - \frac{3\mathbf{a} \cos \mathbf{b}}{\mathbf{a}^2} \right) \\ &= -\sin \mathbf{x} \left(\mathbf{c} - \frac{3 \cos \mathbf{b}}{\mathbf{a}} \right)\end{aligned}$$

...

The loss function of the variational auto-encoder is derived with the following decomposition.

$$\begin{aligned}
L(q, \theta) + KL(q, p) &= \mathbb{E}_q \ln \frac{p(\mathbf{x}, \mathbf{z} | \theta)}{q(\mathbf{z} | \mathbf{x})} - \mathbb{E}_q \ln \frac{p(\mathbf{z} | \mathbf{x}, \theta)}{q(\mathbf{z} | \mathbf{x})} \\
&= \mathbb{E}_q \ln p(\mathbf{x}, \mathbf{z} | \theta) - \mathbb{E}_q \ln q(\mathbf{z} | \mathbf{x}) - \mathbb{E}_q \ln p(\mathbf{z} | \mathbf{x}, \theta) + \mathbb{E}_q \ln q(\mathbf{z} | \mathbf{x}) \\
&= \mathbb{E}_q \ln p(\mathbf{x}, \mathbf{z} | \theta) - \mathbb{E}_q \ln p(\mathbf{z} | \mathbf{x}, \theta) \\
&= \mathbb{E}_q \ln \frac{p(\mathbf{x}, \mathbf{z} | \theta)}{p(\mathbf{z} | \mathbf{x}, \theta)} \\
&= \mathbb{E}_q \ln \frac{p(\mathbf{z} | \mathbf{x}, \theta)p(\mathbf{x} | \theta)}{p(\mathbf{z} | \mathbf{x}, \theta)} \\
&= \mathbb{E}_q \ln p(\mathbf{x} | \theta) \\
&= \ln p(\mathbf{x} | \theta)
\end{aligned}$$

31. What should be in place of $\langle \mathbf{a} \rangle$ and $\langle \mathbf{b} \rangle$?

- A $\langle \mathbf{a} \rangle : p(\mathbf{z}, \mathbf{x} | \theta)p(\mathbf{x} | \theta), \quad \langle \mathbf{b} \rangle : \mathbb{E}_q p(\mathbf{x} | \theta)$
- B $\langle \mathbf{a} \rangle : p(\mathbf{z}, \mathbf{x} | \theta)p(\mathbf{x} | \theta), \quad \langle \mathbf{b} \rangle : \ln p(\mathbf{x} | \theta)$
- C $\langle \mathbf{a} \rangle : p(\mathbf{z} | \mathbf{x}, \theta)p(\mathbf{x} | \theta), \quad \langle \mathbf{b} \rangle : \mathbb{E}_q p(\mathbf{x} | \theta)$
- D $\langle \mathbf{a} \rangle : p(\mathbf{z} | \mathbf{x}, \theta)p(\mathbf{x} | \theta), \quad \langle \mathbf{b} \rangle : \ln p(\mathbf{x} | \theta) \checkmark$

32. Why do we need this decomposition to give us a loss for the VAE?

- A We cannot easily compute $L(q, \theta)$. This provides an upper bound.
- B We cannot easily compute the log probability of $\mathbf{x}$. This provides a lower bound. $\checkmark$
- C This allows us to prove that the VAE converges to the true distribution.
- D This allows us to backpropagate through the sampling step.

See <https://mlvu.github.io/generative/#slide-077>. Note that we prove the decomposition in reverse order: we want but cannot compute $\ln p(\mathbf{x} | \theta)$, and we approximate it by $L + KL$.

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The following dataset represents a spam classification problem: we observe 8 emails and measure two binary features. The first is **True** if the word “crypto” occurs in the e-mail (False otherwise) and the second is **True** if the word “schedule” occurs. The dataset is as follows

“crypto”	“schedule”	label
T	F	Spam
T	F	Ham
F	T	Spam
T	F	Ham
F	F	Spam
F	F	Ham
F	T	Spam
T	T	Ham

We build a naive Bayes classifier on this data, as described in the lecture. We estimate the class priors ($p(\text{Spam})$ and $p(\text{Ham})$) from the data.

33. We observe one email that contains both words and one that contains neither. Which class does the classifier assign to each?
- A both words: Ham, neither word: Ham
 - B both words: Ham, neither word: Spam ✓
 - C both words: Spam, neither word: Ham
 - D both words: Spam, neither word: Spam

The value we're ultimately interested in is

$$p(\text{Class} \mid \text{Email}) \propto p(\text{Email} \mid \text{Class})p(\text{class}) .$$

To figure out which class assigns the highest probability, all we need to do is compute the quantity on the left. If the email contains both words, for the class Ham under the naive Bayes assumption, this becomes:

$$p(C \mid \text{Sp})p(S \mid \text{Sp})p(\text{Sp})$$

Where C stands for the word crypto occurring and $\neg C$ stands for it not occurring. We estimate all required probabilities from the relative frequencies.

$$\begin{array}{ll} p(C \mid \text{Ha}) = \frac{3}{4} & p(\neg C \mid \text{Ha}) = \frac{1}{4} \\ p(S \mid \text{Ha}) = \frac{1}{4} & p(\neg S \mid \text{Ha}) = \frac{3}{4} \\ p(C \mid \text{Sp}) = \frac{1}{4} & p(\neg C \mid \text{Sp}) = \frac{3}{4} \\ p(S \mid \text{Sp}) = \frac{2}{4} & p(\neg S \mid \text{Sp}) = \frac{2}{4} \end{array}$$

It can be helpful not to simplify your fractions ($2/4 \rightarrow 1/2$) for the smoothing question coming up.

The class priors ($p(\text{Sp})$ and $p(\text{Ha})$) we estimate from the data. We have equal amounts of ham and spam, so these are both $1/2$ (also after smoothing).

This gives us for both words occurring:

$$\begin{aligned} p(C | \text{Ha})p(S | \text{Ha})p(\text{Ha}) &= \frac{3}{4} \frac{1}{4} \frac{1}{2} = 3/32 \\ p(C | \text{Sp})p(S | \text{Sp})p(\text{Sp}) &= \frac{1}{4} \frac{2}{4} \frac{1}{2} = 2/32 \end{aligned}$$

So the classification is Ham.

For both words not occurring:

$$\begin{aligned} p(\neg C | \text{Ha})p(\neg S | \text{Ha})p(\text{Ha}) &= \frac{1}{4} \frac{3}{4} \frac{1}{2} = 3/32 \\ p(\neg C | \text{Sp})p(\neg S | \text{Sp})p(\text{Sp}) &= \frac{3}{4} \frac{2}{4} \frac{1}{2} = 6/32 \end{aligned}$$

So the classification is Spam.

34. We observe an email e that contains the word “crypto”, but not the word “schedule”. What probabilities does the classifier assign?
- A $p(\text{Ham} | e) = 9/11$, $p(\text{Spam} | e) = 2/11$ ✓
 - B $p(\text{Ham} | e) = 2/11$, $p(\text{Spam} | e) = 9/11$
 - C $p(\text{Ham} | e) = 9/32$, $p(\text{Spam} | e) = 2/32$
 - D $p(\text{Ham} | e) = 2/32$, $p(\text{Spam} | e) = 9/32$

To get the full probabilities, we need to use the full Bayes’ rule, not the “proportional to” version we used above. However, we can remember that the full Bayes rule is just the quantity $p(\text{Email} | \text{Class})p(\text{Class})$ which we computed in the previous question divided by the sum of that quantity for all classes. So, we first compute

$$p(C | Ha)p(\neg S | Ha)p(Ha) = \frac{3}{4} \frac{3}{4} \frac{1}{2} = 9/32$$

$$p(C | Sp)p(\neg S | Sp)p(Sp) = \frac{1}{4} \frac{2}{4} \frac{1}{2} = 2/32$$

Then we just divide one of these quantities, by the sum of these quantities:

$$p(Ha | Email) = \frac{p(Email | Ha)p(Ha)}{p(Email | Ha)p(Ha) + p(Email | Sp)p(Sp)} .$$

This gives us

$$p(Ha | Email) = \frac{\frac{9}{32}}{\frac{9}{32} + \frac{2}{32}} = \frac{9}{9 + 2} = 9/11$$

$$p(Sp | Email) = \frac{\frac{2}{32}}{\frac{9}{32} + \frac{2}{32}} = \frac{2}{9 + 2} = 2/11$$

The two values should sum to 1, so you could skip the second computation, but it doesn't hurt to have a sanity check.

We add pseudo-observations to the data to deal with unseen emails. The pseudo-observations have the same weight as the normal ones. Compute how the judgments change.

We only need to add enough observations to ensure that every value has been seen once per class per feature. So for **Spam** we add one observation with all F's, one observation with all T's and the same for Ham: four pseudo-observations in total.

“crypto”	“schedule”	label
T	F	Spam
T	F	Ham
F	T	Spam
T	F	Ham
F	F	Spam
F	F	Ham
F	T	Spam
T	T	Ham
T	T	Spam
F	F	Spam
T	T	Ham
F	F	Ham

Alternatively, go back to your original probabilities and add +1 (one pseudo observation) to the numerator and +2 (the number of feature values) to the denominator. Make sure you don’t simplify the fractions if you do this: don’t turn 2/4 into 1/2 and then add the smoothing values. In either case, you should get the following:

$$\begin{aligned}
 p(C | Ha) &= \frac{4}{6} & p(\neg C | Ha) &= \frac{2}{6} \\
 p(S | Ha) &= \frac{2}{6} & p(\neg S | Ha) &= \frac{4}{6} \\
 p(C | Sp) &= \frac{2}{6} & p(\neg C | Sp) &= \frac{4}{6} \\
 p(S | Sp) &= \frac{3}{6} & p(\neg S | Sp) &= \frac{3}{6}
 \end{aligned}$$

This gives us for both words occurring:

$$\begin{aligned}
 p(C | Ha)p(S | Ha)p(Ha) &= \frac{4}{6} \frac{2}{6} \frac{1}{2} = 8/72 \\
 p(C | Sp)p(S | Sp)p(Sp) &= \frac{2}{6} \frac{3}{6} \frac{1}{2} = 6/72
 \end{aligned}$$

So the classification is Ham.

For both words not occurring:

$$p(\neg C | Ha)p(\neg S | Ha)p(Ha) = \frac{2}{6} \frac{4}{6} \frac{1}{2} = 8/72$$

$$p(\neg C | Sp)p(\neg S | Sp)p(Sp) = \frac{4}{6} \frac{3}{6} \frac{1}{2} = 12/72$$

So the classification is **Spam**.

35. We observe one email that contains both words and one that contains neither. Which class does the *smoothed* classifier assign to each?
- A both words: **Ham**, neither word: **Ham**
 - B both words: **Ham**, neither word: **Spam** ✓
 - C both words: **Spam**, neither word: **Ham**
 - D both words: **Spam**, neither word: **Spam**

...

Consider the following task. The aim is to predict the class y from the binary features x_1, x_2, x_3 and x_4 .

x_1	x_2	x_3	x_4	y
A	B	A	B	No
B	A	B	A	Yes
A	A	A	B	Yes
B	B	B	A	No
A	B	B	A	No
B	B	B	A	No
A	A	A	B	Yes
A	A	A	B	Yes
B	B	B	A	No
A	B	B	B	No
B	A	A	A	Yes
B	A	A	B	Yes

36. In standard decision tree learning (as explained in the lectures), without pruning, which would be the first feature chosen for a split?
- A x_1 B x_2 ✓ C x_3 D x_4

37. If we *remove* that feature from the data, which would be chosen instead?

A x_1 B x_2 C x_3 ✓ D x_4

We can work these two questions out by explicitly computing the information gain, but that's a little involved. A shortcut is to apply the *principle* of information gain directly. We want to choose the feature that leads to the *least uniform* class distribution after the split. We can tally up quickly what the class distributions are on both sides of the split for each feature:

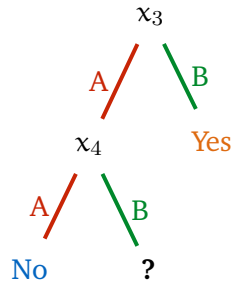
x_1	A	B	x_2	A	B	x_3	A	B	x_4	A	B
Y	3	3	Y	3	3	Y	5	1	Y	2	4
N	3	3	N	3	3	N	1	5	N	4	2

You can sum up the margins of each table to check that your tally is correct (in this case the result is very regular, but don't assume that that is always the case for this question type).

We see that the *least uniform* split is x_3 . That means it's the furthest away from 50/50. If we remove that feature from the data, the next least uniform is x_4 .

NB: In this case, the splits are always the same, whether we choose A or B. If that is not the case, you'll need to compute a weighted entropy. In that case, you can just follow the information gain formula.

With the same data, consider the following (partial) decision tree:



Note that this is just an arbitrary tree, not necessarily corresponding to the choices made in the previous questions.

We can place either x_1 or x_2 on the open node (indicated by the question mark).

38. At this point, what is the information gain for x_1 ?

- A $\frac{3}{4} \log_2(3) - \log_2(5) - \frac{16}{5}$
- B $\frac{3}{4} \log_2(3) + \log_2(5) - \frac{16}{5}$
- C $\frac{3}{5} \log_2(3) - \log_2(5) - \frac{16}{5}$
- D $\frac{3}{5} \log_2(3) + \log_2(5) - \frac{16}{5}$ ✓

Now we have to explicitly compute the information gain. This is the entropy of the class distribution before the split minus the weighted entropy of the class distribution after. Before the split, we have four instances, with labels **YYNN**. That is, a uniform distribution ($p(\text{Y}) = p(\text{N}) = 1/2$), which has an entropy of 1 bit. After the split, we get **N** for **A** (entropy 0 with weight 1/4) and **NYN** for B (entropy $H(1/3)$ with weight 3/4).

Plugging these values into the formula for the gain:

$$\begin{aligned}
 1 - \frac{1}{4} \times 0 - \frac{3}{4} \times H(2/3) &= 1 - \frac{3}{4} \left(-\frac{1}{3} \log \frac{1}{3} - \frac{2}{3} \log \frac{2}{3} \right) \\
 &= 1 - \frac{3}{4} (\log 3 - \frac{2}{3}) \\
 &= 1 - \frac{3}{4} \log 3 - \frac{1}{2}
 \end{aligned}$$

Of course, if you prefer to use a calculator, you can just compute the values in the answer and the values from your own computation.

...

Consider the following function:

$$\mathbf{y} = \mathbf{x}\mathbf{A}$$

Here $\mathbf{x}$ and $\mathbf{y}$ are row vectors and $\mathbf{A}$ is a matrix. Note that this implies that $y_i = \sum_k A_{ki}x_k$.

This function is part of a larger computation graph, resulting in a scalar loss l . We want to implement this computation in an automatic differentiation (AD) system, as discussed in the lectures.

39. Work out the scalar derivative of y_i over x_j . Which is the correct solution?

- A $\frac{\partial y_i}{\partial x_j} = A_{ij}$
- B $\frac{\partial y_i}{\partial x_j} = A_{ji}$ ✓
- C $\frac{\partial y_i}{\partial x_j} = A_{ij}x_i$
- D $\frac{\partial y_i}{\partial x_j} = A_{ji}x_j$

$$\frac{\partial y_i}{\partial x_j} = \frac{\partial \sum_k A_{ki}x_k}{\partial x_j} = \sum_k \frac{\partial A_{ki}x_k}{\partial x_j} = \frac{\partial A_{ji}x_j}{=} A_{ji}.$$

In the middle, we note that there is only one term in the sum that is not zero and that is where $k = j$, so we replace all k 's with j 's and remove the sum symbol.

The AD system will give us a row vector $\mathbf{y}^\nabla$, such that $y_i^\nabla = \frac{\partial l}{\partial y_i}$. Using $\mathbf{y}^\nabla$, we want to efficiently compute a row vector $\mathbf{x}^\nabla$ such that $x_i^\nabla = \frac{\partial l}{\partial x_i}$.

40. Which operation computes this vector for us?

- A $\mathbf{x}^\nabla = \mathbf{x}^\nabla \mathbf{A}^\top$
- B $\mathbf{x}^\nabla = \mathbf{x}^\nabla \mathbf{A}$
- C $\mathbf{x}^\nabla = \mathbf{y}^\nabla \mathbf{A}^\top$
- D $\mathbf{x}^\nabla = \mathbf{y}^\nabla \mathbf{A}$

TODO.

Thank you for your effort. Please check that you've put **your name and student number** on the answer sheet.